RAG Retrieval Enhancement and Model Training Samples

1. Course Content

1. This course introduces the concepts of large model hallucination, RAG retrieval enhancement, and model training samples, laying the theoretical foundation for subsequent practical application and facilitating subsequent understanding and operation.

2. What is large model hallucination? Why do large models experience hallucinations?

RAG retrieval enhancement and model training samples are primarily designed to reduce large model hallucinations. Let's first understand large model hallucinations.

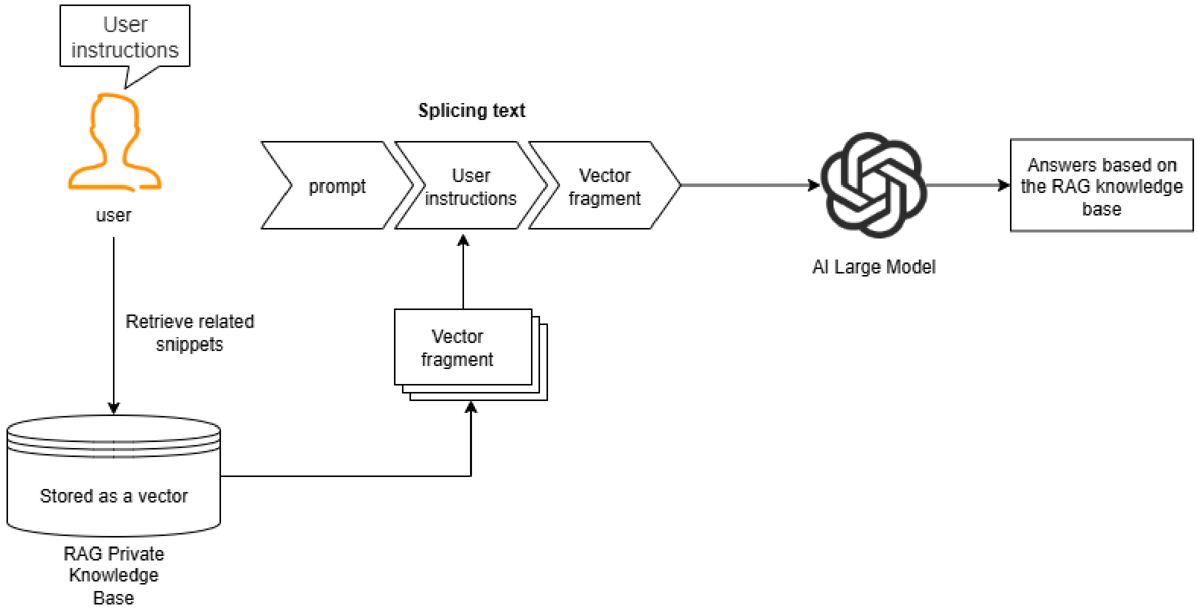
Large model hallucinations refer to situations where the model outputs content that is inconsistent with real-world facts and logic when understanding the environment, generating instructions, or planning actions, resulting in incorrect robot behavior or significant deviations from user expectations.

Why do large model hallucinations occur? **Large models are essentially probabilistic prediction systems trained on massive amounts of text**. In certain domains or scenarios, the training data may lack sufficient samples, causing the model to "fabricate" content to fill gaps.

3. RAG Retrieval Enhancement and Training Examples

3.1 RAG Retrieval Enhancement

RAG is an architecture that combines a retrieval system with a generation model. It aims to address the limitations of traditional generation models in scenarios such as open-domain knowledge question answering and real-time information querying (e.g., knowledge obsolescence, factual hallucinations, and dependencies on long text). Its core logic is to retrieve relevant knowledge through retrieval and incorporate it into prompts, allowing the large model to reference this knowledge, achieving a "retrieval first, generation later" approach.



3.2 Training Examples

Training examples are contained in the RAG knowledge base. The robot comes pre-installed with two knowledge bases: the action function library and the training example library. The training examples include training samples from the course case scenarios, providing insights for the Large model's decision-making and planning in specific scenarios. The subsequent sections [06.AI Model Development] will explain how to configure and extend the proprietary knowledge base and training examples.

4. RAG Retrieval Enhancement and the Role of Training Examples in the Robot

4.1 Reducing the Illusion of a Large Model

When a large model encounters a completely unfamiliar domain or specific user requirements, it will provide possible results based on existing training data. However, these results may differ from the user's expectations or not meet the requirements of a particular scenario.

4.2 Enhancing the Robot's Scene Generalization Ability

The MutoRS robot comes pre-loaded with training examples related to the course cases (explained in detail in the large model configuration course). These training examples provide the large model with reference information in specific scenarios. Users can also add their own training examples to customize their own robot to adapt to different application domains.